ZCAS University

CCS1401 COMPUTING AND ICT SKILLS

MID TERM TEST

21TH APRIL 2026

13:00 – 14: 30 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Any **TWO** questions from three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

Answer any TWO (2) questions.

QUESTION 1 (25 MARKS)

Answer all parts of this question.

a) Define the term utility software. (5 marks)

b) Explain four clear differences between each of the following pairs of Windows 11 utility tools.

I. Disk Defragmentation vs Disk Cleanup (10 Marks).

In your answer, you should bring out differences such as:

- Purpose (what problem each tool solves)
- Type of operation performed on the disk
- Effect on files/data (reorganization vs deletion)
- Impact on system performance

II. System Restore vs File History (10 Marks)

In your answer, you should bring out differences such as:

- What is being restored (system vs personal files)
- When each tool is used
- Impact on user data
- How recovery is performed (restore points vs backups)

QUESTION 2 (25 MARKS)

	A	B	C	D	E	F
1	**REGION**	**ITEMS SOLD**	**ITEM PRICE**	**TOTAL PRICE (ZMW)**	**TOTAL PRICE (US$)**	**TARGET MET**
2	Northern	239	ZMW 2,259.41			
3	Southern	781	ZMW 460.95			
4	Eastern	560	ZMW 500.00			
5	Western	609	ZMW 623.97			
6	Southern	1200	ZMW 266.67			
7	Northern	126	ZMW 3,650.79			
8	Southern	160	ZMW 3,125.00			
9	Eastern	3200	ZMW 62.50			
10	Western	1600	ZMW 250.00			
11	Eastern	320	ZMW 750.00			
12	Northern	640	ZMW 1,187.50			
13	Southern	128	ZMW 3,562.50			
14	Eastern	256	ZMW 1,171.88			
15	Western	120	ZMW 3,416.67			
16	Northern	300	ZMW 1,666.67			
17						
18	**Count of All Regions**	?				
19	**Average of Items Sold**	?				
20	**Lowest Price Per Item**	?				
21	**Highest number of items sold**	?				
22	**Count of Total Price (US$) below $10,000**	?				
23						
24		**US$ Rate**	**29.35**			

For the above spreadsheet, write the formula to find the following: **(You are required to write the formulae the way they would appear in excel i.e using the excel syntax)**

1. Calculate the total price (ZMW) (2 marks)
2. Calculate the total price (us$), exchange rate in b24 (2 marks)
3. Find count of total price (US$) below $10,000 (4 marks)
4. Calculate if the target has been met or not TARGET MET (If TOTAL PRICE (ZMW) > 400000 then the target is "ACHIEVED", otherwise it is "NOT ACHIEVED". (4 marks)
5. Write a formular to calculate the total amount of money in Kwacha made from the southern region. (5 marks)
6. Find Count of Periods, Average of Items Sold, Lowest Item Price and Highest number of Items Sold. (8 marks)

(25 Marks)

QUESTION 3 (25 MARKS)

a) The following storage media are listed: Hard Disk Drive (HDD), Solid State Drive (SSD), Magnetic Tape and Optical Disc (CD/DVD). (10 marks)

Complete the table by filling in the most appropriate features for each storage device. (Draw the table in your answer booklet to answer)

Feature	HDD	SSD	Magnetic Tape	Optical Disc (CD/DVD)
Technology Used				
Speed				
Portability				
Storage Capacity				
Typical Use				

Feature Guides

1. Speed (Use ONLY these terms):
   Very Fast, Fast, Moderate, Slow, Very Slow
2. Technology to store data (Be specific)
3. Portability (Use comparative terms):
   - High (e.g easy to carry, small size) OR Medium because…,
4. Storage Capacity (Use relative terms):
   - Very High (enterprise-level, large-scale storage)
   - High (hundreds of GBs to TBs)
   - Moderate (tens to hundreds of GBs)

- Low (limited capacity)

5. Typical Use (Be specific, not vague)

b) Using your knowledge of computer hardware, match each hardware component below with the most appropriate function/description. (15 marks)

Functions/Descriptions

i. Manages communication between the processor, memory, and peripheral devices at the motherboard level
ii. Provides a pathway for data, control signals, and addresses within the computer system
iii. Allows additional hardware components to be added to enhance system capabilities
iv. Stores low-level instructions essential for hardware initialization and startup
v. Retains system configuration settings such as date, time, and hardware parameters using a small battery
vi. Temporarily stores frequently used instructions to reduce access time from main memory
vii. Handles parallel processing tasks, especially for rendering complex graphical data
viii. Converts alternating current into regulated direct current required by internal components
ix. Enables digital communication between a computer and external networks
x. Uses non-volatile memory with no moving parts to improve data access speed and durability

Hardware Components

1. Cache Memory
2. Chipset
3. Expansion Slot
4. Firmware (BIOS/UEFI)
5. CMOS
6. System Bus
7. Solid State Drive (SSD)
8. GPU
9. Power Supply Unit (PSU)
10. Network Interface Card (NIC)

Instructions.

- Write your answers in the format: i: SSD
- Each correct match = 1.5 marks × 10 = 15 marks

END OF EXAMINATION